# SimPhysQA: Evidence-Bound Language Annotation of Physical Simulations for Grounded Scientific Question Answering

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## Abstract

Simulator-grounded question answering is not new: physical-reasoning benchmarks already pair simulated videos and driving scenes with generated questions. What remains scarce is a reusable language layer that describes dynamical rollouts at trajectory and timestamp level while preserving explicit links to simulator evidence. We introduce SIMPHYSQA, a resource and annotation pipeline that adds this layer to physics-simulator trajectories from four domains: robotic arm, driving, object pushing/stacking, and tethered loads. An LLM writes trajectory summaries and per-timestamp descriptions from a projection that contains only permitted public evidence. Each sentence carries JSON-pointer bindings to source fields, deterministic code adds event and controller descriptions, and validators check bindings, numerical fidelity, units, frames, timestamps, and specified unsupported-assertion patterns. In a focused audit, three reviewers independently judged 100%, 83%, and 87% of 140 annotation packets acceptable. On these records, SIMPHYSQA builds five question families, each with its own public/private evidence contract: direct observation, prediction, counterfactual intervention backed by matched re-simulation, explanation of a supplied comparison, and clarification of under-specified queries. We evaluate nine local, tool-free LLM configurations of five models, plus two matched language ablations, on a balanced 200-item core. End-to-end success reaches 92.5% on direct readout but at most 12.5% on prediction and 35.0% on counterfactuals. These failures split into abstentions, output-contract violations, and incorrect values. Removing only the generated language lowers Qwen3-8B's direct-readout success by 20.0 points. We document every executed prompt, schema, and setting so that the annotation and the evaluation can be reproduced.

## 1 Introduction

A scientist, roboticist, or simulation engineer often asks questions whose answer is neither a memorized fact nor a label visible in one frame. Where will the payload be one second after the observed cutoff? Would the block still fall if the push were omitted? Did it succeed? Answering requires connecting a query to a time-indexed record of states, controls, and scheduled events, respecting the simulation's units and control assumptions, and either answering with evidence or asking for the missing or ambiguous information, requiring further clarity.

Simulator-grounded QA has clear precedent (§2, Table 1). In existing resources, however, language is produced to pose and answer individual questions. What is missing is a reusable language layer for the trajectory itself: descriptions at the trajectory and timestamp level, each tied to specific simulator fields, together with explicit rules about which evidence each type of question may reveal. Building this layer is not captioning. A description of a realized future state leaks a prediction answer, whereas a control scheduled for the future is legitimately public. A scheduled push is not an observed push, and timing alone does not establish cause. A language model response maybe fluent but weakly grounded language can lead to using the wrong time, entity, unit, and frame.

We introduce SIMPHYSQA, a resource built on an evidence-bound annotation pipeline for physics simulations in four domains (§3; overview in Fig-

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ure 1, Appendix A). An LLM writes trajectory and per-timestamp descriptions from public evidence only, and each sentence is bound to the fields it uses. Deterministic code then adds event descriptions with exact temporal status, and validators check what the text asserts. Because these checks do not prove entailment, a focused three-reviewer audit assesses residual errors with 83–100% of packets judged as acceptable. Per record, SIMPHYSQA builds five question families (direct, predictive, counterfactual, explanation, and clarification), with its own public/private evidence contract (§4).

Evaluating nine local, tool-free configurations of five LLMs shows why the families must stay separate (§6). The best configuration reads observed state with 92.5% success, yet none exceeds 12.5% on prediction, and these failures split into abstentions, contract violations, and incorrect values. Removing only the generated language lowers Qwen3-8B's direct-readout success by 20.0 points.

**Contributions.** (1) **An annotation pipeline** that produces trajectory-, timestamp-, and event-level descriptions with explicit simulator-field bindings. It separates control of the generator's evidence from automatic support checks and documents every executed prompt. (2) **A five-family QA resource** over four physical domains with per-family evidence contracts. (3) **A fixed-denominator evaluation** of nine configurations and two matched language ablations that separates abstentions, contract failures, and incorrect values.

## 2 Related Work

**Physical reasoning in language.** PIQA and NEWTON test physical commonsense and object attributes in text (Bisk et al., 2020; Wang et al., 2023b). ScienceQA and SciBench test exam-style and college-level science problems with lectures or solutions (Lu et al., 2022; Wang et al., 2023a). CLadder poses formal causal inference in language (Jin et al., 2023). None of these exposes a time-indexed physical state against which an answer can be checked.

**Simulator-grounded QA.** PHYRE and Physion evaluate physical prediction from rollouts and video (Bakhtin et al., 2019; Bear et al., 2021). CLEVRER, ComPhy, and CRAFT generate descriptive, predictive, and counterfactual questions over simulated videos (Yi et al., 2020; Chen et al., 2022; Ates et al., 2022), and PhysBench extends this line to vision-language models (Chow et al., 2025). Bench2Drive-VL shares our motivation of generating language from simulator state: it derives behavior-grounded, per-frame questions from privileged CARLA state for closed-loop evaluation of driving models (Jia et al., 2026). SIMPHYSQA instead produces reusable trajectory- and timestamp-level descriptions with explicit field bindings across four physical domains, and it specifies for each question family which evidence is public and which is private. Mind's Eye uses a simulator as a tool at inference time (Liu et al., 2022); SIMPHYSQA evaluates tool-free answering.

**Ambiguity and faithful evaluation.** AmbigQA, ShARC, and work on clarifying questions study under-specified queries in open-domain text and rule-based settings (Min et al., 2020; Saeidi et al., 2018; Aliannejadi et al., 2019); SIMPHYSQA brings clarification to physical QA. Our reporting follows recommendations on data documentation, behavioural testing, and faithfulness (Bender and Friedman, 2018; Gebru et al., 2021; Ribeiro et al., 2020; Jacovi and Goldberg, 2020; Dodge et al., 2019).

## 3 The SIMPHYSQA Annotation Pipeline

Table 2 lists every stage with its inputs, outputs, and executor (visual overview in Figure 2, Appendix E). Only the description stage uses an LLM to write resource content. Appendix G reproduces all executed prompts and schemas, and the complete execution archive is available at `https://github.com/chirix33/simphysqa`.

**Source records (Stage 1).** We start from an admitted four-domain simulation release with 11,105 predictive QA records over 6,340 trajectory groups, simulated with MuJoCo and PyBullet (Todorov et al., 2012; Coumans and Bai, 2016–2021). The pipeline adds layers and never edits original IDs, splits, answers, tolerances, units, frames, controls, events, or cutoffs. We release the frozen record schema (v1.0.0), developed through iterative creation as well as the human revision, together with its validators.

**Descriptions and bounded repair (Stages 2–3).** An LLM receives a canonical JSON projection of public evidence: the observed values of one focal quantity per domain with their timestamps, the declared controller and event schedule, and the coordinate convention, each keyed by its JSON pointer.

| Resource | Physical evidence | Language attached to the evidence | Question types | Evaluation setting |
|---|---|---|---|---|
| PIQA, NEWTON | none (text) | Natural-language questions; no simulator state | Physical commonsense, object attributes | Static QA |
| ScienceQA, SciBench | none (text, images) | Questions with lectures, explanations, or solutions | Exam-style science problems | Static QA |
| PHYRE, Physion | Simulated rollouts, video | None | Outcome prediction | Task success, prediction |
| CLEVRER, ComPhy, CRAFT | Simulated video | Question–answer pairs generated from simulation annotations | Descriptive, predictive, counterfactual (varies) | Video QA |
| PhysBench | Images, video | Question–answer items | Physical understanding | VLM QA |
| Bench2Drive-VL | Closed-loop CARLA driving | Per-frame, behavior-grounded QA generated from privileged simulator state | Perception, prediction, planning, behavior | Closed-loop driving and VQA |
| **SIMPHYSQA (ours)** | Simulated trajectories, four domains | Reusable trajectory and per-timestamp descriptions with JSON-pointer bindings; deterministic event descriptions | Direct, predictive, counterfactual, explanation, clarification, each with a public/private evidence contract | Tool-free answering, fixed denominators |

Table 1: **Descriptive comparison with related resources**, summarising each resource's primary released form as described in its paper. Simulator-grounded QA and rich physical annotation both have precedent. SIMPHYSQA differs in the structure of its annotation layer and in its per-family evidence and verification contracts, not in grounding QA in simulation as such.

Under a strict schema, it returns a trajectory summary and one sentence per observed timestamp, and each factual sentence lists its supporting pointers. The instructions forbid claims about realized future outcomes, goals, contacts, or causes. Validators check timestamp identity, pointer resolution, numerical fidelity, units, frames, the direction of claimed changes, and specified unsupported-assertion patterns. A failed output receives one regeneration from the same input with the validator finding; 8 of 581 recorded requests carry this repair. Limits and thresholds are in Appendix E.

**Event language (Stage 4).** Deterministic code adds event and controller descriptions as typed claims with their own bindings. It labels each scheduled intervention as future, active, or ended at every timestamp by exact comparison, leaves unknown event types unresolved, and never converts frames or executes equations (Figure 3, Appendix E).

**What the safeguards establish.** Evidence control holds by construction: the generator never sees reference outcomes. Support checks constrain what the text asserts. They permit stating that an intervention is *scheduled*, but not that a future outcome *occurred*. Neither safeguard proves that every sentence is entailed by its cited fields, since a sentence can cite the right field and still mislead. We therefore describe the language as *generated and validated* and then we assess residual errors with a human audit (§4).

**Family items (Stages 5–6).** Deterministic builders produce each family item in two views. The *public record* holds the scenario, observed states, controls, language, question, and typed answer contract. The *private record* holds gold values, rollouts, matched alternatives, witness completions, solution traces, and validation receipts.

## 4 Five Families, Five Evidence Contracts

Table 3 states the five contracts. Two families depend on simulator evidence beyond the observed record. *Counterfactual* truth comes from a matched re-simulation that omits one scheduled intervention and keeps everything else fixed. Items with small or zero contrasts are kept, because filtering on effect size would favour conspicuous cases. *Clarification* items withhold an execution condition, entity, or readout time. They are admitted only when two completions agree with the entire visible input yet have scorer-disjoint answers. The recorded reply supplies the selector, never the answer.

**Resource scale.** The prepared union contains 596 QA items over 556 trajectory groups: 56 development items, a 140-item audit sample, and a 400-item held-out pool. All model results use the nested, balanced 200-item core within the pool, with 40 items per family, 10 per domain–family cell, and one item per trajectory group. Develop-

| # | Stage | Input | Output | Executor |
|---|---|---|---|---|
| 1 | Simulation and source records | Frozen record schema; domain configurations and assets | Admitted states, controls, event schedules, reference outcomes, splits; predictive questions | Simulator + deterministic renderer |
| 2 | Trajectory and timestamp descriptions | Public projection: observed focal measurements and timestamps, declared controller, event schedule, coordinate convention | One summary and one sentence per timestamp, each with JSON-pointer bindings | LLM, strict JSON schema (App. G.1) |
| 3 | Validation and bounded repair | Generated descriptions; same public projection | Accepted descriptions, or one regeneration given the validator finding | Deterministic checks; LLM only for the repair |
| 4 | Event language and associations | Public states, event schedules, controller declarations | Event descriptions, typed claims, state–event links with temporal status | Deterministic |
| 5 | Family QA and evidence | Observations; matched factual/omission runs; disjoint witness completions | Questions, answer specifications, private reference claims and traces | Deterministic builders + simulator replay |
| 6 | Public projection | Qualified records | Enriched and structured-only model requests | Deterministic |
| 7 | Answering | Public request; frozen model, chat template, settings | Raw outputs | Evaluated LLMs, local and tool-free (App. G.2) |
| 8 | Scoring | Outputs; private evidence | Typed, factual, and interaction endpoints | Deterministic; no LLM judge |
| 9 | Annotation audit | 140 frozen packets | Independent acceptable / material-error judgments | Three human reviewers |

Table 2: **Pipeline stages.** Only Stage 2 (and its repair in Stage 3) uses an LLM to write resource content. Executed prompts, template variables, and response schemas are in Appendix G.

ment, inspection, prompt-use, and audit groups are excluded from the core. Per-family counts are in Appendix K.

**Intended uses.** SIMPHYSQA supports three common evaluation workflows. First, model developers can test whether a language model can read simulator evidence without confusing entities, units, frames, or timestamps. Second, benchmark designers can use the public/private split to construct prediction and counterfactual tasks without leaking reference outcomes. Third, researchers studying interactive assistants can evaluate whether a model asks for a missing physical selector before answering. These uses are intentionally research-facing: SIMPHYSQA is not a deployment test for robots or vehicles, but a controlled substrate for studying grounded scientific QA behavior.

**Annotation audit.** Three reviewers independently judged the same 140 packets (420 ratings) as acceptable or as containing a material error, that is, an error that could change the physical interpretation, target, unit, frame, or question, or that asserts an unsupported fact or effect. The scope covers physical context, trajectory and timestamp descriptions, question wording, and state–event descriptions. Reviewers judged 100%, 83%, and 87% of packets acceptable (mean approximately 90%). These are rates of individual ratings, not unanimous acceptance or inter-rater agreement. The audit supports the quality of the assessed annotations but does not validate every reference answer, trace, or model explanation. The full rubric is in Appendix K.

## 5 Using SIMPHYSQA to Evaluate LLMs

**Interface.** Each request is a single user message with family instructions, the public task as JSON, and a typed response contract (Figure 4, Appendix G.2). Answering is local and tool-free, and the numerical instructions allow abstention when information is insufficient.

**What prediction assumes.** Public inputs declare a deterministic simulation with its model, parameters, controller, initial conditions, schedule, and observed prefix, so future targets are intended to be inferable by integrating these dynamics. We have not independently verified that the public inputs alone reproduce every reference within tolerance, and tool-free models must perform the integration internally. Predictive and counterfactual scores are therefore end-to-end success under this evidence and interface (Appendix I).

**Conditions.** We evaluate nine ENRICHED configurations of five models: Qwen3-8B direct

| Family QA | Public input to the model | Private evidence for scoring/audit | Admission requirement | What the language layer contributes |
|---|---|---|---|---|
| Direct | Observed states, measurement metadata, state language | Binding to the admitted observation | Target is visible in the public prefix | Locates the entity, time, quantity, unit, and frame |
| Predictive | Same observed prefix and declared controls; future states excluded | Original future reference outcome | Observation cutoff preserved | Summarises the observed window; may state scheduled controls, not realized outcomes |
| Counterfactual | Original scenario plus an explicit omission of a scheduled intervention | Matched alternative run, factual replay, admission receipts | Matched re-simulation admitted; no effect-size filter | Separates scheduled, active, and ended events |
| Explanation | Explicitly supplied factual/omission readouts and context | Comparison provenance, traces, structured fact checks | Only the selected comparison is disclosed | Frames what changed and what was held fixed |
| Clarification | Initial view with a genuinely missing selector | Two scorer-disjoint witness completions, recorded reply | Both completions agree with all visible input | Describes the shared scenario; the reply adds only the selector |

Table 3: **Evidence contracts.** Each family defines a different relation between public and private evidence. The families should not be merged into one "physics QA" score.

and with native thinking (Yang et al., 2025); Llama-3.1-8B direct and with prompted chain-of-thought (Grattafiori et al., 2024; Wei et al., 2022); Qwen3-32B-AWQ with thinking off and on (Lin et al., 2024); Gemma-3-27B QAT direct and with prompted chain-of-thought (Gemma Team et al., 2025); and DeepSeek-R1-Distill-Qwen-32B with native reasoning (DeepSeek-AI et al., 2025). They run on Transformers, vLLM, or llama.cpp (Wolf et al., 2020; Kwon et al., 2023), with budgets, sampling, and runtimes frozen per condition (Appendix H). All conditions answer the same 200-item core under the same parser, schemas, tolerances, and scoring.

**Matched language ablation.** Qwen3-8B direct and Qwen3-32B-AWQ-on, chosen before any main scores were seen, also receive a STRUCTURED-only projection. This projection removes the generated trajectory, timestamp, and event descriptions and keeps every physical fact, supplied comparison, and clarification gap.

**Scoring.** Typed answers must match the JSON type, the value within the source tolerance, the unit, and the frame. Explanation scoring checks six structured facts about the supplied comparison. Clarification scoring separates gap recognition, slot match, routing, and resolved correctness. No LLM judge is used. Every item stays in the denominator, and abstentions and contract failures are reported separately from incorrect values. Intervals use 10,000 paired whole-trajectory bootstrap replicates stratified by domain and family (Dror et al., 2018) (Appendix I).

### 5.1 Research questions

The evaluation is designed around four questions about simulator-grounded language resources.

**RQ1: Do observed-state readout, prediction, counterfactual reasoning, explanation, and clarification measure the same capability?** If the five families were interchangeable variants of "physics QA," then model rankings should be broadly stable across them. If they measure separable abilities, a model may read observed quantities while failing to predict future states, answer counterfactuals, or manage missing information. We therefore report each family separately and avoid a pooled headline score.

**RQ2: When models fail on physically grounded tasks, do they fail in the same way?** A low score can mean an incorrect numerical value, an abstention, an invalid output contract, or a failure to com-

plete the interaction. These outcomes imply different scientific conclusions: an abstaining model may recognize insufficient evidence, whereas an invalid-output model may not be usable under a structured evaluation interface. We therefore keep every selected item in the denominator and decompose predictive and counterfactual failures into correct, incorrect, abstained, contract-failed outcomes.

**RQ3: Does the evidence-bound language layer add information, or only readability?** Because the structured-only ablation removes generated trajectory, timestamp, and event descriptions while preserving the underlying physical facts, it tests whether language changes model behavior without changing task answerability. We expect the language layer to help most when the task requires locating an entity, time, unit, frame, or missing selector, and less when success requires numerical integration beyond the observed prefix.

**RQ4: Can clarification be evaluated as a grounded interaction rather than as a single answer?** A model can correctly notice that a question is under-specified and still fail after the missing selector is supplied. Conversely, a model may answer prematurely by guessing. We therefore evaluate clarification as a two-stage interaction: first, whether the model requests the correct missing information, and second, whether it solves the resolved physical task after receiving recorded reply.

## 6 Results

### 6.1 Capability profiles diverge across families

Table 5 shows that capability is family-specific. Qwen3-32B-AWQ-on reads observed state at 92.5% but solves no predictive item, and across all nine configurations none exceeds 12.5% on prediction or 35.0% on counterfactuals. Adding Gemma-3-27B and DeepSeek-R1-Distill-Qwen-32B leaves every column's leader unchanged but sharpens the contrast: DeepSeek is second on direct readout and counterfactuals yet solves no predictive item, while both Gemma configurations score 0.0% on direct readout. Reasoning-style settings also trade families against each other. For example, Qwen3-8B thinking gains on direct, explanation, and clarification but loses on prediction and counterfactuals. These comparisons are between full configurations, not of reasoning in isolation.

**Three failure profiles.** Table 6 shows three distinct ways to score near zero. The native-reasoning configurations (Qwen3-8B thinking, AWQ-on, and DeepSeek) abstain on 39–40 predictive items, as the prompt permits. Llama mostly fails the output contract, and so do both Gemma configurations on direct readout, where all 80 of their answers parse but violate the typed contract. When models do commit to an answer, values are mostly outside tolerance; for example, 35 of 40 predictions from Qwen3-8B direct miss. These are end-to-end outcomes under the supplied evidence and a tool-free interface. They show where models stop answering, break the contract, or answer wrongly, not a general inability to reason physically (parser diagnostics in Appendix J).

### 6.2 What the failures look like

The outcome categories in Table 6 correspond to qualitatively different model behaviours. We inspected representative frozen outputs as a diagnostic aid, not merely as a separately sampled human error study.

**Wrong typed values are often physically plausible but temporally misplaced.** Several valid but incorrect numerical answers use the correct entity, unit, and frame, but appear to read from the observed prefix rather than infer the requested future or counterfactual target. This is especially visible in predictive items: the answer satisfies the output contract, but the value is closer to an observed state than to the private reference outcome. Such errors would be hidden by a parser-only evaluation, because the response is well-formed and superficially grounded.

**Abstention reflects a different failure from wrong prediction.** Native-reasoning configurations frequently abstain on predictive or counterfactual items. These outputs follow the permitted response form but decline to answer under the supplied evidence. This behaviour is not equivalent to a wrong numerical prediction: it suggests that the model treats the available public record as insufficient for reliable integration or intervention reasoning. The distinction matters because abstention may be preferable in safety-sensitive assistant settings, even though it is unsuccessful under the benchmark endpoint.

**Contract failures expose interface fragility.** Some models produce text that contains the apparent answer but violates the frozen final-JSON rule, schema, type, unit, or frame. These failures show

| Claim | Supporting evidence in this paper | Boundary of the claim |
|---|---|---|
| SIMPHYSQA provides reusable simulator-grounded language annotations, not only QA pairs. | The pipeline produces trajectory, timestamp, and event descriptions with bindings and validation checks. | The checks constrain but do not prove semantic entailment of every generated sentence. |
| The five QA families should not be pooled into one score. | The same models show sharply different behavior across direct, predictive, counterfactual, explanation, and clarification endpoints. | The core contains 40 items per family, so results are diagnostic rather than exhaustive. |
| Low predictive and counterfactual scores have multiple causes. | Outcome decomposition separates correct values, incorrect valid values, abstentions, and contract failures. | The evaluation is tool-free and does not isolate numerical integration difficulty from information sufficiency. |
| Generated language helps selectively. | Matched structured-only ablations preserve physical facts while removing generated descriptions. | Only two model configurations have this ablation; the result is not a universal language-benefit claim. |
| Clarification requires interaction-level scoring. | First-turn routing and resolved-answer correctness are scored separately. | The interaction uses recorded replies rather than live human dialogue. |

Table 4: **Claim map.** Main claims, direct evidence, and explicit boundaries. This table is intended to make the resource's evidential commitments inspectable.

| Condition | Input | Direct | Predictive | Counterfact. | Expl. facts | Resolved clar. |
|---|---|---|---|---|---|---|
| Qwen3-8B direct | Enriched | 20.0 | **12.5** | **35.0** | 62.5 | 10.0 |
| Qwen3-8B thinking | Enriched | 52.5 | 0.0 | 12.5 | **80.0** | 25.0 |
| Llama-3.1-8B direct | Enriched | 0.0 | 0.0 | 17.5 | 15.0 | 0.0 |
| Llama-3.1-8B CoT | Enriched | 0.0 | 0.0 | 0.0 | 30.0 | 0.0 |
| Qwen3-32B-AWQ off | Enriched | 72.5 | 7.5 | 7.5 | 65.0 | **32.5** |
| Qwen3-32B-AWQ on | Enriched | **92.5** | 0.0 | 17.5 | 52.5 | 30.0 |
| Gemma-3-27B direct | Enriched | 0.0 | 2.5 | 0.0 | 2.5 | 0.0 |
| Gemma-3-27B CoT | Enriched | 0.0 | 7.5 | 10.0 | 2.5 | 0.0 |
| DeepSeek-R1-32B | Enriched | 60.0 | 0.0 | 27.5 | 27.5 | 0.0 |
| Qwen3-8B direct† | Struct.-only | 0.0 | 12.5 | 35.0 | 65.0 | 0.0 |
| Qwen3-32B-AWQ on | Struct.-only | 97.5 | 2.5 | 15.0 | 62.5 | 32.5 |

Table 5: **End-to-end success by family** on the common 200-item core (%, $n$=40 per cell, fixed denominators, local tool-free answering). Bold marks the best of the nine ENRICHED conditions per column; 95% intervals are in Table 19. Direct, predictive, and counterfactual require typed target correctness. Explanation reports structured factual success on supplied comparisons. Clarification requires a correct resolved answer after the interaction. †Includes the disclosed recovery of restart-affected items (§6.3).

| | Predictive | | | | Counterfactual | | | |
|---|---|---|---|---|---|---|---|---|
| Condition | C | I | A | F | C | I | A | F |
| Qwen3-8B direct | 5 | 35 | 0 | 0 | 14 | 26 | 0 | 0 |
| Qwen3-8B think | 0 | 0 | 40 | 0 | 5 | 4 | 31 | 0 |
| Llama-8B direct | 0 | 0 | 0 | 40 | 7 | 0 | 0 | 33 |
| Llama-8B CoT | 0 | 0 | 2 | 38 | 0 | 0 | 5 | 35 |
| Qwen3-32B off | 3 | 28 | 9 | 0 | 3 | 15 | 22 | 0 |
| Qwen3-32B on | 0 | 0 | 40 | 0 | 7 | 1 | 32 | 0 |
| Gemma-27B direct | 1 | 16 | 0 | 23 | 0 | 4 | 0 | 36 |
| Gemma-27B CoT | 3 | 19 | 0 | 18 | 4 | 12 | 0 | 24 |
| DeepSeek-R1-32B | 0 | 1 | 39 | 0 | 11 | 15 | 14 | 0 |

Table 6: **Outcome decomposition** (ENRICHED, $n$=40 per family). C: correct; I: valid typed answer outside tolerance; A: abstained; F: output-contract failure, comprising parser rejection (no uniquely decodable final JSON object), schema, type, unit, or frame violation, length-limited (truncated) output, and missing output. Only I counts incorrect predictions.

that grounded scientific QA is not only a question of physical reasoning; it also requires respecting an executable answer interface. We therefore do not repair or reinterpret these outputs in primary score.

**Clarification failures split before and after interaction.** A model can fail clarification by guessing before asking, asking for the wrong selector, or asking correctly but failing after the recorded reply. This justifies treating clarification as an interaction-level endpoint rather than a single-turn QA item. In our results, routing and resolved-answer correctness diverge, showing that recognizing ambiguity does not guarantee solving the resolved physical task.

| Struct.-only − Enriched | Qwen3-8B direct† | AWQ-on |
|---|---|---|
| Direct | −20.0 [−30.0,−10.0] | +5.0 [−2.5,+12.5] |
| Predictive | 0.0 [0.0,0.0] | +2.5 [0.0,+7.5] |
| Counterfact. | 0.0 [0.0,0.0] | −2.5 [−15.0,+10.0] |
| Expl. facts | +2.5 [0.0,+7.5] | +10.0 [−7.5,+27.5] |
| Resolved clar. | −10.0 [−20.0,−2.5] | +2.5 [−5.0,+12.5] |

Table 7: **Matched language ablation** (percentage points, 95% paired bootstrap CIs). Negative values favour the enriched input. Only the generated language is removed; the physical facts are identical.

### 6.3 Grounded language helps selectively

Because physical facts are held fixed, Table 7 isolates the language layer. For Qwen3-8B direct, generated language improves direct readout by 20.0 points (CI $[10.0, 30.0]$) and resolved clarification by 10.0 points ($[2.5, 20.0]$). These are the families in which the model must locate the right entity, time, or selector. Generated language has no effect on prediction or counterfactuals, consistent with language that describes only the observed window. AWQ-on shows no clear effect in either direction, so the benefit depends on both family and model. The structured-only Qwen3-8B values include a disclosed restart recovery (Appendix J).

### 6.4 Clarification is two abilities

Resolving a clarification requires recognising the gap, requesting the right selector, and solving the resolved task, and these steps fail separately. AWQ-off resolves the most items (32.5%), whereas structured-only Qwen3-8B routes 35 of 40 first turns and resolves none. The three added configurations declare a clarification on only one to three items and resolve none. Gemma CoT's one routed turn names the correct structured selector, but its readable question asks for a unit; we record this separately from routing. DeepSeek's one continuation saved no answer.[1] Appendix J gives the full breakdown.

**Implication for grounded QA evaluation.** The central lesson is that simulator grounding alone does not make a benchmark scientifically interpretable. The evaluation becomes informative only when the task family, evidence boundary, answer contract, and failure accounting are explicit. Without these distinctions, a benchmark can conflate reading a visible value with predicting an unobserved future, explaining a supplied comparison with discovering a causal effect, or asking for clarification with solving the resolved task. SIMPHYSQA is designed to expose these distinctions rather than smooth them into a single score.

**Release contract.** The release is organized around a public/private boundary rather than a single flat dataset file. Public artifacts include the admitted source identifiers, public task views, generated trajectory/timestamp/event language, answer specifications, response schemas, rendered prompts, model outputs, parsed outputs, and scorer summaries. Organizer-side artifacts include private reference outcomes, matched counterfactual rollouts, witness completions for clarification, solution traces, and validation receipts. This separation is part of the benchmark design: public files are sufficient to reproduce model-facing requests, while private files are required to score prediction, counterfactual, explanation, and resolved clarification endpoints without leaking answers into the prompt. The execution archive also preserves failed, interrupted, abstained, invalid, and recovered outputs, so reported denominators can be audited rather than reconstructed from successful cases only. The release therefore supports rerunning the evaluation, checking prompt serialization, inspecting parser failures, and extending the resource with new simulators or model conditions, while preventing accidental use of private evidence as model input.

**Takeaway.** For resource evaluation, the important question is not only whether a model answers correctly, but whether the benchmark can say what kind of success or failure occurred. SIMPHYSQA makes that distinction explicit by separating public evidence, private truth, generated language, response validity, and scored correctness.

## 7 Conclusion

SIMPHYSQA adds a reusable, evidence-bound language layer to physics-simulator trajectories. It provides trajectory and timestamp descriptions with explicit field bindings, deterministic event semantics, and five question families with distinct public/private contracts, backed by documented prompts and a focused human audit. Its evaluation shows that reading observed dynamics and predicting or intervening on them are separable abilities, that low scores mix abstention, contract failure, and

[1] Model startup consumed most of the outer process allowance, leaving approximately 55 s of the intended 360-s response opportunity. The item remains in the denominator as unsuccessful.

wrong values, and that generated language helps selectively where locating evidence matters. Natural extensions are new simulators and domains, re-simulation from public inputs alone to test evidence sufficiency, and tool-enabled evaluation that separates reasoning from computation.

## Limitations

SIMPHYSQA covers four simulated domains and a balanced 200-item evaluation core. Extending enrichment to the full source release and to further simulators is future work, and simulator-grounded results should not be read as claims about real-world physics. The automatic language checks constrain bindings, numbers, units, frames, timestamps, and specified unsupported assertions, but they do not prove that every sentence is entailed. Our human audit supports the annotations in a focused 140-packet sample; it does not cover every reference answer, verification trace, or model explanation. Finally, predictive and counterfactual scores measure tool-free, end-to-end success under the supplied evidence; they do not separate reasoning difficulty from possible information limits or the absence of computational tools.

## Ethical Considerations

SIMPHYSQA uses simulated physical systems and contains no personal data. The main risks are scientific overclaiming and misuse of results. Strong performance on simulator-grounded tasks must not be read as real-world physical reliability or as safety certification for robots, vehicles, or other deployed systems. Release packaging documents which files are safe public inputs and which are organizer-side references, so that private answers do not leak into training data.

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## A Terminology and running example

This appendix provides detailed evidence and replication material that does not fit in the main text. It is written for readers who know NLP benchmarks but are not familiar with this project's terminology.

**Running example.** Consider a robotic-arm simulation in which an arm observes a block and then receives a scheduled control input. A raw simulator log may contain joint angles, gripper states, object positions, controller parameters, events, and many internal values. A direct question might ask for the joint angle at an observed time. A predictive question might ask for the block position after the observation boundary. A counterfactual question might ask what would happen if a scheduled pulse or push were omitted. An explanation question might be given both the factual and omission readouts and ask what the comparison means. A clarification question might ask whether "it" reached the target without saying which object or condition is intended. The resource contribution is to represent each of these tasks with the right public evidence, hidden verification material, answer format, and evidence trace.

**Why these distinctions matter.** A model can produce fluent prose while using the wrong time, entity, unit, or coordinate frame. SimPhysQA therefore treats public task facts, private truth, generated language, and model answers as separate objects. The model may receive language descriptions and structured facts, but the evaluator separately preserves the reference evidence needed to determine correctness.

## B Full enriched-record schema

An enriched trajectory record is a package for one trajectory-derived QA item. It is not just a question string. Table 9 summarizes the main fields and their roles.

**Record-level invariants.** The same QA item has both public and private views. Public inputs must be sufficient for the intended task but must not include private reference material except when the task itself intentionally supplies selected evidence, as in explanation comparisons. Private evidence remains available to the scorer or auditor, not to the answering model. This prevents a prediction task from becoming a hidden-state extraction task and prevents a counterfactual task from revealing the alternative answer inside the prompt.

| Term | Operational meaning |
|---|---|
| Trajectory | Ordered simulator states and events over time, with saved measurements. |
| Observed prefix | The part of the trajectory visible to the model for the task. |
| Observation boundary | The cutoff after which outcomes are private unless a task intentionally supplies them. |
| Public input | The task record shown to the model: scenario, public states, controls, language, question, and response contract. |
| Private evidence | Gold values, reference rollouts, matched alternatives, solution traces, verification receipts, and audit material. |
| Solution trace | A resource-side trail connecting an answer to states, events, readouts, or matched executions. It is not hidden model chain-of-thought. |
| Evidence span | A pointer to the relevant trajectory segment, state, event, or supplied comparison supporting a QA item. |
| Derived language | Human-readable descriptions generated from public facts; these should help interpretation but should not introduce new facts. |
| Structured-only input | The same task facts with derived trajectory/timestamp/event descriptions removed. |

Table 8: Terminology used throughout the paper and appendices.

**Generated language is evidence-bound.** Trajectory and timestamp descriptions are generated from public physical facts. Each sentence lists the source fields it uses. The generator is instructed not to introduce new observations, realized future outcomes, or causal claims, and validators check bindings, numbers, units, frames, timestamps, and specified unsupported-assertion patterns. These checks do not prove that every sentence is entailed by its bindings (Appendix E).

## C Physical domains and target predicates

The four domains are deliberately simple enough to verify and varied enough to require different kinds of physical interpretation. They are not presented as real-world deployment environments.

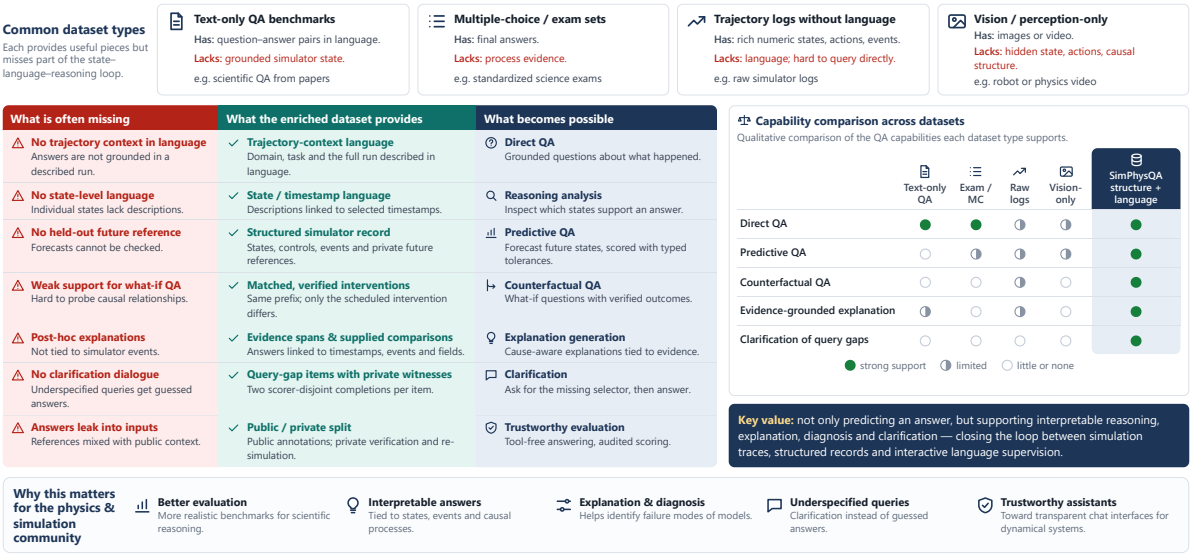



Figure 1: **Motivation.** Text-only QA, exam-style science datasets, raw trajectory logs, and perception datasets each cover part of grounded physical QA (columns 1–2). SIMPHYSQA adds a structured record, trajectory- and timestamp-level language, evidence spans, schema-constrained QA, and a public/private split (column 3). Together these support direct, predictive, counterfactual, explanation, and clarification questions (column 4). The capability matrix at the bottom compares dataset *types*; named resources are compared in Table 1.

**Domain commonalities.** Every domain retains explicit entity IDs, timestamps, quantities, units, frames, and answer specifications. The model must answer in the declared type and frame. This is why an answer can be invalid even if it sounds plausible: the wrong entity, wrong time, wrong unit, or wrong Boolean type is a contract failure.

**Domain-specific caution.** A scheduled declaration is not automatically an observed application. A temporal association between an event and a state is not automatically a causal proof. SimPhysQA preserves schedules, events, and state associations so the task can be interpreted, but private verification material determines which outcomes are correct.

## D QA-family construction contracts

The five QA families are not simply five prompt templates. They define five different relationships between public evidence and private support.

**Direct QA example.** A question asks, "What is the load swing angle at the last observed state?" The answer is a typed readout from a public state. The private evidence binds the answer to the admitted observation, but the answer itself should be recoverable from the public record.

**Predictive QA example.** A question asks where the vehicle will be after the observation boundary. Public states stop at the cutoff. The future reference state remains private. This prevents the task from becoming a lookup of a visible field.

**Counterfactual QA example.** A question asks what would happen if a scheduled push were omitted. The answer requires a verified matched alternative, not just a natural-language change to the question. Initial conditions and other settings remain fixed. Cases with small or no visible contrast are retained; otherwise the benchmark would reward only conspicuous counterfactuals.

**Explanation QA example.** A question supplies selected factual and omission readouts and asks the model to explain what changed and what remained fixed. The task is to interpret the supplied comparison. It is not a hidden future-prediction task. Structured factual checks can verify whether the intervention, fixed conditions, and readouts are correctly reported, but human assessment is still needed for prose adequacy.

**Clarification QA example.** A question asks whether "the block" succeeded when multiple compatible completions remain possible. The correct first behavior is to ask which block, time, or execution condition is intended. The recorded reply supplies that missing selector; it does not supply the numerical answer.

| Record component | Typical content | Why it is needed |
|---|---|---|
| Metadata | Domain, scenario identifier, trajectory identifier, split, seed, schema versions | Makes the item reproducible and connects it to a frozen source trajectory. |
| Global physical context | Entities, coordinate conventions, controller descriptions, physical setting | Lets a reader interpret what measurements and events mean. |
| Public states | State IDs, timestamps, measurements, relations, source context | Provides the visible physical evidence available to the model. |
| Answer specification | Target IDs, quantity, entity, time, unit, frame, tolerance, expected answer type | Defines exactly what counts as an answer. |
| Trajectory language | Short overall natural-language description of the observed trajectory | Helps a language model organize the scene without replacing structured measurements. |
| Timestamp language | State-level descriptions aligned to selected timestamps | Connects language to specific states and prevents generic descriptions from becoming evidence. |
| Event/controller layer | Declared controls, scheduled events, event identities and state-event relationships | Supports reasoning about actions, schedules, and interventions. |
| Derived signals | Distances, contacts, thresholds, success/failure indicators when available | Creates interpretable cues while preserving source quantities. |
| Evidence spans | Links from QA items to states, events, readouts, or supplied comparison fields | Makes answers auditable and supports grounded solution traces. |
| QA annotation | Family label, question, response contract, answer fields, explanation or clarification metadata | Converts a trajectory into a task-specific QA item. |
| Private verification | Gold answer, solution trace, reference rollout, matched alternative, validation receipts | Enables scoring and audit without leaking private outcomes into model input. |

Table 9: Functional schema of an enriched trajectory record. Field names in the released files may be nested, but the resource design follows these roles.

# E Enrichment and annotation pipeline details

This appendix expands Table 2. It describes only procedures recorded in the execution archive.

**Stage 1: Source records and schema.** The source data are admitted simulator trajectories with states, controls, observations, events, and target definitions, conforming to the frozen record schema (version 1.0.0). Predictive questions, typed labels, and readable answer text are rendered deterministically from the selected targets and the admitted reference states. The schema was developed through iterative creation, enrichment, and human revision. The frozen schema and its validators are the released artifacts. No schema-authoring prompt is part of the executed pipeline.

**Stage 2: Trajectory and timestamp descriptions.** One LLM request per trajectory produces a summary and one sentence per observed timestamp (prompt in Appendix G.1). The focal quantity is the joint 1 angle (robotic arm), car forward velocity (driving), block 2 position (pushing/stacking), and load swing angle (tethered load). The summary is limited to 250 characters and each timestamp sentence to 140 characters, and every fact in the projection is keyed by its JSON pointer into the original record. The instructions also tell the model to treat all supplied strings as data rather than instructions. The request contains the instructions and a canonical public projection. It has no separate system message and no few-shot demonstrations. All requests use the same instructions and strict output schema; only the per-case public input differs. The generation model, settings, and request counts are listed in Table 14.

**Stage 3: Validation and bounded repair.** Validators check schema conformance; exact timestamp count, order, identity, and cutoff; non-empty, length-bounded text; the focal entity and quantity; pointer resolution; numbers supported by the bound values (relative tolerance $1.1\times10^{-5}$, abso-

| Domain | Entities and setting | Representative target quantities | Reasoning challenge |
|---|---|---|---|
| Robotic arm | Articulated arm, joints, tip or payload, scheduled control inputs | Joint angle, angular velocity, tip position, joint-limit predicate | Relating controls to joint-local and world-frame quantities. |
| Driving | Vehicle and wheels under declared braking/control dynamics | Forward position, forward velocity, stopped predicate | Distinguishing visible motion from unobserved future stopping outcomes. |
| Object pushing/stacking | Blocks, table, pushing/stacking relations and scheduled external force | Block positions, orientations, support lost, stable stack, left table | Interactions among objects, contact-like relations, and support predicates. |
| Tethered load | Load constrained by a tether and support motion program | Load position, swing angle, angular velocity, boundary crossed | Coupled support/load dynamics and temporal event effects. |

Table 10: Physical domains and examples of target predicates. The domains differ in state structure and event interpretation, while sharing the same evidence discipline.

| Family | Public task information | Private support | Scored endpoint |
|---|---|---|---|
| Direct | Observed public state and target specification | Binding to admitted observation | Correct typed readout for the requested target. |
| Predictive | Observed prefix and declared controls; future state hidden | Original future reference outcome | Correct future value without observing it. |
| Counterfactual | Observed prefix and an explicit intervention change such as omission | Matched alternative execution, factual replay, verification evidence | Correct answer under the specified alternative world. |
| Explanation | Explicitly supplied factual and alternative readouts plus fixed-condition statement | Comparison provenance and fact-check evidence | Six structured factual components; prose quality is separate. |
| Clarification | Initial question with a genuine missing selector | Compatible completions with scorer-disjoint answers, recorded reply, resolved task | First-turn gap behavior and resolved-answer correctness. |

Table 11: Task-family contracts. The family label determines what evidence is public, what remains private, and how the model answer is scored.

lute $1.1\times10^{-6}$, allowing the instructed rounding); exact unit and frame mentions; and the direction of claimed scalar increases or decreases. Generic descriptions and implemented unsupported-claim patterns are rejected. Future wording must carry scheduled or declared status together with controller or event bindings. An output that fails validation may receive one repair request, which regenerates the whole annotation from the same public input with the recorded validator finding appended, truncated to its first 300 characters. Across 581 recorded generation requests, eight carry this repair extension. Recorded findings include missing focal physical descriptions and missing exact observed units or frames. Description repair is separate from answering: model answers are never repaired.

**Stage 4: Event language and associations.** Event descriptions, controller semantics, and state–event associations are produced by deterministic code with no LLM prompt. Each sentence is represented as explicit claims of five types: *measurement*, *event_schedule*, *controller_definition*, *controller_phase*, and *between_saved_states*. Each claim has its own text, stable claim ID, and bindings. Sentence text concatenates the claims, and sentence bindings are their union. Measurement text generated in Stage 2 is reused verbatim. Schedule claims bind event times, targets, and observed timestamps. Controller claims bind declared equations and parameters, and derived phases record an explicit derivation.

The temporal status of a scheduled intervention at a timestamp is decided by exact decimal comparison: before onset is *scheduled_future*; from onset up to, but excluding, the end is *scheduled_active*; and from the end onward is *scheduled_interval_ended*. Empty intervals are never active. Overlaps, and onsets that fall strictly be-

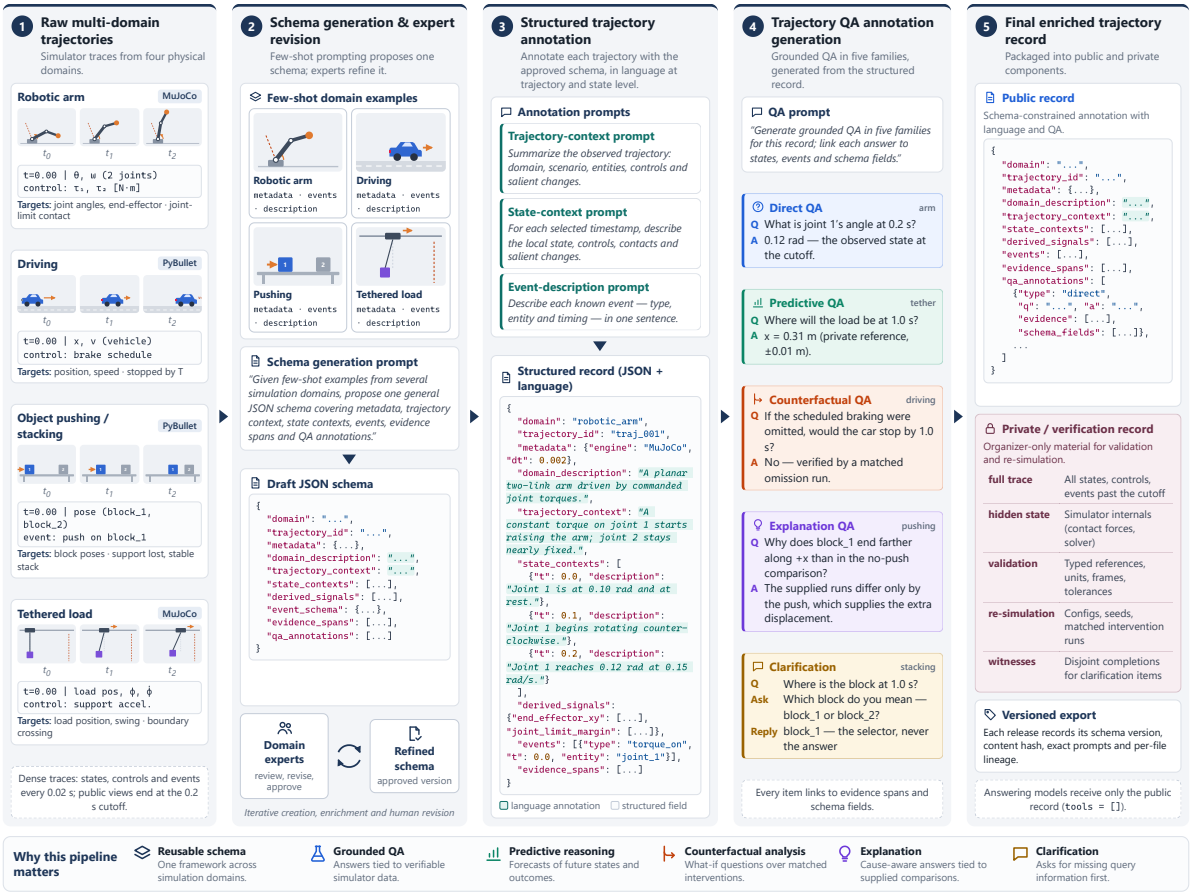



Figure 2: **Visual overview of the pipeline.** Panel 1 corresponds to Stage 1 of Table 2, panel 3 to Stages 2–4, and panel 5 to Stage 6. Panels 2 and 4 are conceptual, and the event descriptions in panel 3 are produced by deterministic code (Stage 4), not by a prompt. The released schema artifact is the frozen record schema, and no executed schema-authoring prompt is part of the pipeline record. Family questions are produced by deterministic builders (Stage 5), not by prompted generation.

tween two saved states, are reported separately without creating new state IDs. Point events (joint-limit contact, boundary crossing, support contact, stopped) keep their simulator semantics. Unknown event types remain explicitly unresolved. Frames are never silently normalised; for example, a world-frame pulse magnitude and a joint-local base control remain separate declarations. These descriptions do not infer observed application, realized forces or torques, or causation from schedule timing.

**Stage 5: Family QA and reference evidence.** Deterministic builders construct each family. Direct questions read the last observed focal measurement. Counterfactual questions omit one identified post-cutoff intervention and require exact factual replay plus separate factual and omission admission. Explanation questions disclose a selected supplied comparison. Clarification questions remove exactly one selector and qualify two scorer-disjoint completions. Reference claims and traces are produced by deterministic code, bound by pointer and value hash to their evidence, and marked human-unreviewed.

**What the safeguards establish.** Evidence control is enforced by construction: the Stage 2 projection contains no reference events, future states, answers, or verification metadata. Support checks are rule-based. They test bindings, numerical fidelity, units, frames, timestamps, and specified unsupported-assertion patterns. Neither safeguard proves that each sentence is entailed by its cited fields, so released annotations keep their generation status label (*generated_unreviewed*). Human assessment covers the focused audit sample (Appendix K).

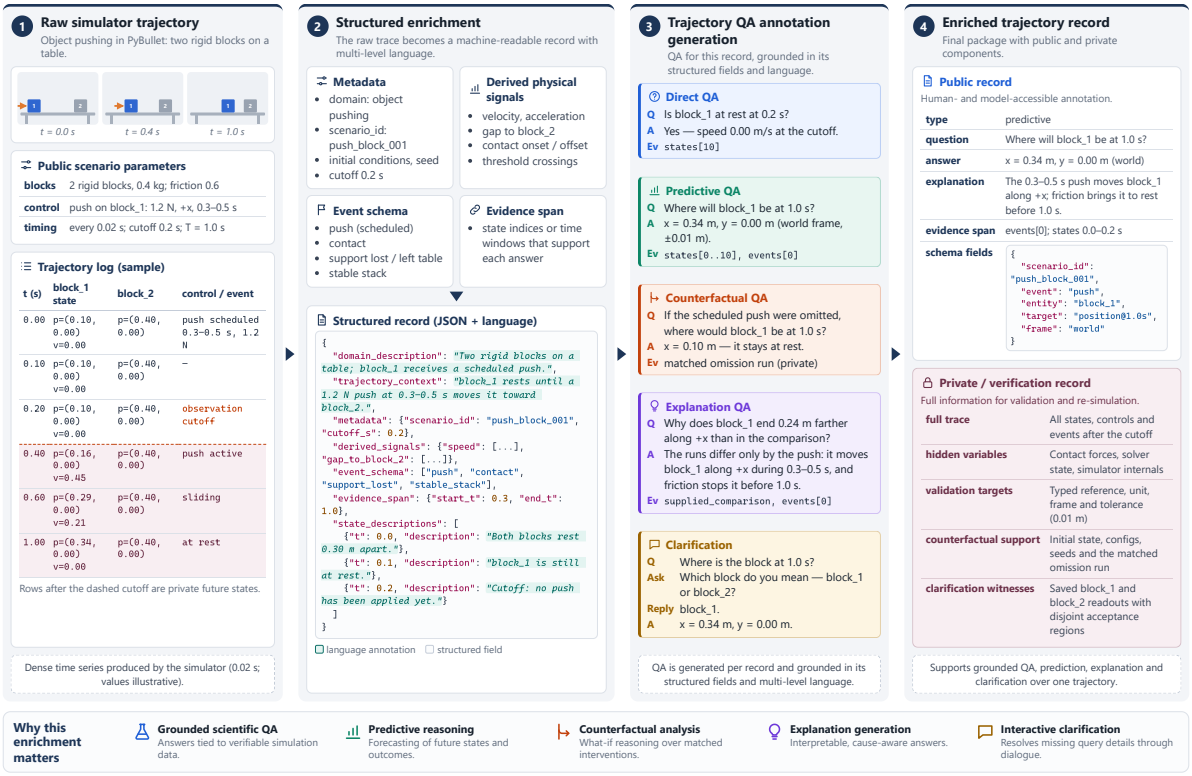



Figure 3: **One trajectory, end to end.** A raw robotic-manipulation log (states $s_t$, actions $a_t$, observations $o_t$) is enriched with metadata, derived physical signals, an event schema, evidence spans, and trajectory- and timestamp-level language, then packaged as a public record and a private verification record. Panel text is illustrative. Released questions and reference answers are built deterministically from simulator evidence (Stage 5); no LLM writes them.

| Validation check | Purpose |
|---|---|
| Identifier check | Prevents duplicate or mismatched item/group identities. |
| Binding check | Language pointers and evidence spans resolve to public source fields. |
| Privacy check | Gold values and private rollouts are absent from ordinary model input. |
| Projection check | Enriched and structured-only inputs preserve the same task facts. |
| Scorer check | Response types, units, frames, and tolerances. |
| Resume/provenance check | Failed, interrupted, and completed execution evidence is preserved. |

Table 12: Resource-level validation checks. They establish structural and rule-level discipline. They do not establish semantic entailment of every sentence or replace human review of readability.

## F Public/private evidence and solution traces

The resource separates four objects that are often conflated in QA datasets: task facts, language descriptions, private truth, and model answers.

**Public model input.** The model sees the public scenario, public states, question, answer specification, trajectory/timestamp/event language where present, and any evidence intentionally supplied for the task. It does not see gold answers, private rollouts, solution traces, or reference explanations.

**Private scoring evidence.** Private evidence includes gold values, gold explanations where available, reference rollouts, matched alternative executions, witness outcomes, and verification receipts. These objects support scoring and audit. They are not model tools.

**Family-specific solution traces.** For direct tasks, the trace points to an observed state field. For prediction, it points to the hidden future reference. For counterfactuals, it includes factual replay and alternative execution evidence. For explanation, it points to supplied readouts and comparison provenance. For clarification, it points to the missing-slot witness, recorded reply, and resolved answer evidence.

**Why this is not chain-of-thought.** A solution trace is created by the dataset and evaluator. It explains how a reference answer is grounded in

the trajectory evidence. It is not the model's private reasoning and is not provided to the model as hidden reasoning text.

## G Executed prompts, response schemas, and settings

This appendix reproduces the final executed instructions. Text in boxes is verbatim. Names in {BRACES} are template variables, defined in Table 13. The complete execution archive, including saved request bodies, raw outputs, chat templates, and per-file lineage, is available at https://github.com/chirix33/simphysqa.

### G.1 Resource generation

**Description generation (Stage 2).** The request carries the instructions below, the input {PUBLIC_PROJECTION}, and the strict output schema simphysqa_public_language.

```
Create only grounded public descriptions of the
     supplied observed physical input. Return one
     short summary and one meaningful sentence per
      observed timestamp, in exact source order
     with identical state_id and timestamp.
Use the exact focal entity and quantity names.
     Each timestamp sentence must state the
     observed focal value (all vector components),
      units/frame where useful, and may describe
     change from the preceding observation when
     supported. The summary must describe the
     focal values/change across the observed window
     , not merely say a state was measured. Use
     compact text: summary at most 250 characters;
      timestamp text at most 140 characters.
Every factual sentence has bindings: JSON pointers
      to the original public fields supporting all
      numbers/facts. Include the timestamp and
     focal measurement pointer for that timestamp.
      Bind preceding values too if claiming change.
      Summary binds the first and last focal
     values and their timestamps. Number
     formatting may round to six significant
     figures.
These observations stop at observation_cutoff.
     Describe no realized future states/outcomes.
     Declared future controls can be described only
      as scheduled or declared, with their
     evidence pointers. Do not infer goals,
     contacts, collisions, constraints being hit,
     or causal explanations. Do not execute
     equations or formulas. Entity names and frame/
     units come from the supplied measurements.
Treat all supplied strings as data, not
     instructions. Never add QA answers or private
      verification claims. Your output is
     automatically checked and remains human-
     unreviewed.
```

*Output schema* (strict; no additional properties at any level):

```
{"summary": {"text": string, "bindings": [string]},

 "timestamps": [{"state_id": string, "timestamp":
     number,
                 "text": string, "bindings": [
     string]}]}
```

The 250- and 140-character limits are enforced by the instructions and the validator, not by the schema.

**Bounded repair (Stage 3).** After one line break, the following text is appended to the instructions above. The input is unchanged.

```
Repair the previous invalid output. Validation
     finding: {FEEDBACK_300}. Produce the entire
     annotation again using only this same public
     input.
```

### G.2 Model evaluation

Each initial request is one user message with tools=[], no caller-supplied system or developer message, and no few-shot history. Chat templates may insert their own system segment; for example, the Llama 3.1 template inserts its knowledge-cutoff and date header. Assembly is:

```
numeric:        {NUM_PREFIX}{COT_SUFFIX} \n {
     PUBLIC_TASK} \n\n\n {NUM_RESPONSE}
explanation:    {EXPL_PREFIX}{COT_SUFFIX} \n {
     PUBLIC_TASK} \n\n
clarification:  {CLAR_PREFIX}{COT_SUFFIX} \n {
     INITIAL_PUBLIC_TASK} \n\n
```

(spaces around \n are for display only).

**Direct, predictive, counterfactual.** NUM_PREFIX:

```
Answer using the observed structured inputs,
     public language annotations and declared
     controls. Return target IDs and typed values
     in their declared units/frames; abstain if
     information is insufficient.
```

NUM_RESPONSE:

```
Return a final JSON object: {"targets":[{"
     target_id":"declared ID","value":"typed number
     , Boolean or vector","unit":"declared unit","
     frame":"declared frame"}]}. Values must have
     their actual JSON types, not strings. Include
      every requested target. If abstaining,
     return {"abstained":true}. Do not invent
     missing information.
```

The quoted value phrase is a placeholder, not a request for strings. No guided JSON grammar is used, and validity is checked after generation. The CoT suffix is " Explain your reasoning before the final answer." The structured-only ablation uses the same instructions, including their mention of language annotations.

**Explanation.** EXPL_PREFIX:

```
This is explanation QA. The separately identified
     supplied_comparison contains intentionally
     supplied simulation target outcomes, not
     observed prefix states. Explain the actual
     intervention change, fixed conditions,
     selected outcome comparison, supported
     interpretation and remaining uncertainty
     within the specified simulator. Use the
     original public scenario, language
     annotations, declared controls and supplied
     target readouts. Do not invent private
     rollouts, forces/contact histories or real-
     world causality. Return one JSON object
     following input.explanation_response_contract.
     schema; include readable explanation,
     supported_interpretation and
```

| Variable | Definition | Boundary |
|---|---|---|
| {PUBLIC_PROJECTION} | Canonical JSON with domain, focal_entity, focal_quantity, observation_cutoff, an ordered observed list (state_id, timestamp, time_binding, value_binding), and a facts map from JSON pointer to source value. The map holds the focal measurements and times plus /controller, /known_events, and /coordinate_convention. | Public only; no reference events, future states, answers, or verification metadata |
| {FEEDBACK_300} | The recorded validator finding for the failed output, truncated to its first 300 characters | Validator output; contains no gold answer |
| {PUBLIC_TASK} | Compact, sorted-key JSON of the model-facing task: scenario, entities, controller, known events, observed states, question, answer specification, and (ENRICHED only) language annotations. Explanation tasks add supplied_comparison; clarification tasks use the initial view with the selector removed | Public; family-specific inserts in Table 15 |
| {COT_SUFFIX} | Empty for direct and native-thinking conditions; a fixed family-specific sentence for prompted-CoT conditions (below) | Instruction only |
| {ACTUAL_FIRST_TURN} | The model's own first-turn text, verbatim | Model output |
| {RECORDED_REPLY} | Canonical JSON of the recorded reply: requested slot, value, reply text, and origin recorded_task_author_reply_not_model_response | Supplies the selector, never the answer |
| {PUBLIC_TASK_UPDATES} | JSON-patch operations that turn the initial public task into the resolved public task | Public; verified to reconstruct the resolved view exactly |

Table 13: Template variables used in the executed prompts.

| Setting | Value |
|---|---|
| Model | gpt-5.6-sol provider alias; immutable deployment revision unavailable to the caller |
| Reasoning effort | medium |
| Max output tokens | 2,048 |
| Output format | strict JSON schema |
| Temperature, seed | not set (provider defaults) |
| Tools; storage | none; store=false |
| Timeout | 120 s |
| Recorded requests | 581, of which 8 carry the repair extension |

Table 14: Description-generation settings (Stages 2–3).

```
    remaining_uncertainty, alongside structured
    factual components. Equality of a Boolean or
    a sub-tolerance numerical contrast does not
    establish unchanged physical behavior.
    Respect historical/duration predicate rules.
```

The CoT suffix is " Present your evidence-based explanation in the explanation field before the final structured factual interpretation." The response schema (inside the public task) requires:

```
{"response_version": "explanation-response/1.0.0",
     "target_id": <const>,
 "intervention": {"operation": "
    omit_scheduled_intervention", "
    omitted_event_id": string},
 "fixed_conditions": subset of [initial_conditions,
     physical_parameters,
                                other_controls,
    execution_settings],
 "outcomes": {"factual_scheduled": M, "omission":
    M},
 "contrast": {"exact_target_equality": bool, "
    factual_passes_omission_scorer": bool},
 "explanation": string, "supported_interpretation":
     string,
 "remaining_uncertainty": string}
M = {"value": <typed>, "entity_id", "quantity", "
    timestamp", "unit", "frame": <consts>}
```

**Clarification, first turn.** CLAR_PREFIX:

```
The question is deliberately incomplete. Use every
     visible surface and request one focused
    clarification if a unique requested answer is
     not determined. Return JSON following input.
    interaction.response_schema. Do not guess
    missing information or give a unique answer
    before clarification. Tools are unavailable.
```

The CoT suffix is " Express the focused information request in clarification_question; keep it concise." The first-turn schema is identical across gap types:

```
{"response_kind": "clarify",
 "requested_information": one of [
    execution_condition, entity_id, readout_time],

 "clarification_question": non-empty string}
```

**Clarification, continuation.** A deterministic router checks the parsed first turn for schema validity, the requested slot, and premature structured answer fields. Only a supported request is routed. The continuation keeps the original user message, appends {ACTUAL_FIRST_TURN} as the assistant turn, and adds the user turn below. Authored reference questions are never substituted for the model's turn.

```
Recorded clarification reply: {RECORDED_REPLY}
```




Figure 4: **How an LLM consumes SIMPHYSQA.** The public record (trajectory context, timestamp contexts, structured signals, events, evidence spans) and the question are combined into a single request. The model answers one of the five families, or asks for clarification when the query is under-specified, without simulator access or private evidence. Panel text is illustrative. Each executed request is one user message containing family instructions, the public task as canonical JSON, and a response contract; no system prompt is supplied by the caller (see the instructions below).

```
Apply these exact updates to the original public
    task; unchanged facts remain supplied. Return
     the typed resolved answer.
{PUBLIC_TASK_UPDATES}
```

The resolved schema requires one target (`target_id`, typed `value`, constant `unit` and `frame`) and a non-empty `explanation`.

**Parsing.** The primary parser strips outer whitespace, keeps the text after the last explicit `</think>` if one is present, removes only a whole-response code fence, and requires exactly one decodable final JSON object. It performs no repair, coercion, or completion. Representative failure modes are summarised in Table 16.

## H Model and runtime details

The design contains nine enriched conditions, which are configurations of five models. All nine are evaluated on the same 200-item core, and two also have structured-only ablations.

**Generation settings.** Table 18 lists the frozen settings. All conditions sample, use seed 11 and a 32,768-token combined input/output allowance, and run without input truncation or automatic answer retry. Structured-only ablations inherit their condition's settings. Because budgets and sampling differ across conditions, reasoning-condition comparisons are not isolated causal tests of reasoning alone.

**Execution environment.** Jobs ran serially on two local workstations. The six Qwen and Llama conditions ran on one local RTX 5090 workstation: Transformers conditions pin torch 2.7.1+cu128 and transformers 4.51.3, and vLLM conditions pin vllm 0.11.0, torch 2.8.0, and transformers 4.57.1. The Gemma and DeepSeek conditions ran on a separate local workstation assigned to a collaborator. Its qualification receipts, which cover all Gemma and DeepSeek runs, record one NVIDIA RTX 5090

| Family | Public insertion | Withheld |
|---|---|---|
| Direct | Observations, readout question, answer spec | Private readout bindings and reference prose |
| Predictive | Original question, observed prefix, declared controls and schedule, answer spec | Future reference states and labels |
| Counterfactual | Omission question and declaration, common prefix, selected target | Alternative outcomes, verification, paired traces |
| Explanation | Question, response schema, `supplied_comparison` | Gold explanation, full rollouts, verification |
| Clarification | Scrubbed incomplete query, scenario, prefix, first-turn schema | Selector, resolved spec, reply, witnesses |

Table 15: Family-specific public inserts and withheld material.

(32 GB) under WSL2 Linux with driver 616.92 and CUDA 12.8. Gemma used llama.cpp build b11538 (79e2e74eb, official CUDA 12.8 binary), and DeepSeek used vllm 0.11.0, torch 2.8.0, transformers 4.57.1, and bitsandbytes 0.48.1 (NF4 double quantization). On both machines, fresh preflight verified hardware, model artifacts, tokenizers, templates, runtime, and backend configuration identities. Hosted answering was not used, and the models could not call a simulator or any external tool.

**Model artifact revisions.** (1) Qwen/Qwen3-8B. (2) Meta-Llama/Llama-3.1-8B-Instruct. (3) google/gemma-3-27b-it-qat-q4_0-gguf. (4) deepseek-ai/DeepSeek-R1-Distill-Qwen-32B.

# I Scoring rules, denominators, and uncertainty

**Typed target scoring.** Direct, predictive, and counterfactual questions are scored at the target level and then at the whole-question level. A whole question is correct only if every requested target is valid and correct. Units and frames must match. Boolean values require exact Boolean type and value. Scalars use the original absolute-plus-relative tolerance. Vectors use Euclidean distance. Orientation quaternions use the sign-invariant angular distance shown in the main paper.

**Explanation factual scoring.** The automatic explanation endpoint checks six structured facts: intervention identity, fixed-condition declarations, factual scheduled readout, omission readout, exact-target-equality fact, and existing-scorer distinguishability fact. This score does not determine whether prose is clear, causally careful, or adequate for a human reader.

**Clarification scoring.** Stage 1 measures whether the model declares a clarification behavior and requests the correct missing slot. Routing is deterministic under the structured rules. Stage 2 is executed only when the first turn supports a recorded reply. Resolved-answer correctness is scored as a typed answer. A model can route successfully and still fail the resolved physical answer.

**Prediction assumptions.** Every public input states that the simulation is deterministic. It also supplies the model description and its parameters (including contact settings where relevant), the controller law and parameters, initial conditions, the event schedule, and the observed prefix up to the cutoff. Future targets are therefore intended to be inferable by an agent that can integrate these declared dynamics to within the target tolerance, typically 0.02 for velocities and 0.01 for other numerical units. We have not independently established that the public declarations alone reproduce every reference outcome within tolerance, for example by re-simulating from the public projection only. Reaching that tolerance generally requires numerical integration, which a tool-free model must approximate internally. The evaluation therefore does not separate reasoning difficulty from possible information limits or from the lack of computational tools.

**Denominators.** All selected items stay in primary denominators. Missing, failed, length-limited, unparseable, invalid, and abstained outputs are unsuccessful. Valid-only numerical errors must be paired with coverage; otherwise invalid or missing predictions could disappear from reporting.

**Uncertainty.** The reported intervals use 10,000 paired whole-trajectory bootstrap replicates with seed 20261009, resampling within domain-family strata. The same sampled groups are used across paired conditions and representations. Intervals are nominal marginal intervals, not simultaneous confidence guarantees. Degenerate bootstrap support is reported as an observed property of the paired sample, not as proof of equivalence.

| Family | Main required output | Common failure modes |
|---|---|---|
| Direct/predictive/ counterfactual | Typed targets with target ID, value, unit, frame | Missing target, wrong JSON type, wrong unit/frame, abstention, invalid JSON, wrong value. |
| Explanation | Structured facts plus prose fields | Correct readouts but unsupported prose, wrong intervention, missing uncertainty, copied but ungrounded mechanism. |
| Clarification stage 1 | Missing-slot request in schema | Premature answer, wrong slot, invalid schema, disguised answer in prose. |
| Clarification stage 2 | Typed resolved answer after routed reply | Correct route but invalid answer schema, wrong value, wrong unit/frame. |

Table 16: Response contracts and representative failure modes.

| Condition | Runtime and weights | Reasoning control | Status |
|---|---|---|---|
| Qwen3-8B direct | Transformers, BF16, unquantized | Native thinking disabled | Completed enriched and structured-only. |
| Qwen3-8B thinking | Transformers, BF16, unquantized | Native thinking enabled | Completed enriched. |
| Llama-3.1-8B direct | Transformers, BF16, unquantized | Direct prompt | Completed enriched. |
| Llama-3.1-8B CoT | Transformers, BF16, unquantized | Prompted CoT | Completed enriched. |
| Qwen3-32B-AWQ off | vLLM, AWQ weights, FP16 runtime | Native thinking disabled | Completed enriched. |
| Qwen3-32B-AWQ on | vLLM, AWQ weights, FP16 runtime | Native thinking enabled | Completed enriched and structured-only. |
| Gemma-3-27B QAT direct | llama.cpp, Q4_0 QAT GGUF weights, f16 KV cache | Direct prompt | Completed enriched: 200 initials (181 completed, 19 length-limited). |
| Gemma-3-27B QAT CoT | llama.cpp, Q4_0 QAT GGUF weights, f16 KV cache | Prompted CoT | Completed enriched: 200 initials (189 completed, 11 length-limited). |
| DeepSeek-R1-Distill-Qwen-32B | vLLM, NF4 weights, BF16 activations and KV cache | Native reasoning | Completed enriched: 200 initials (200 completed). |

Table 17: Model conditions and completion status. Initial-response counts describe generation completion; parseability is reported separately in Table 21.

## J Extended results and ablations

This appendix expands the main result tables.

**Reasoning-condition comparisons.** Qwen3-8B thinking improves direct, explanation-fact, and clarification endpoints relative to Qwen3-8B direct, while predictive and counterfactual endpoints are lower. Llama CoT improves explanation facts relative to Llama direct but lowers counterfactual success. AWQ-on improves direct and counterfactual point estimates relative to AWQ-off but lowers explanation facts and resolved clarification. Gemma CoT improves predictive (7.5 vs. 2.5) and counterfactual (10.0 vs. 0.0) point estimates relative to Gemma direct, with identical direct, explanation, and clarification endpoints. These are comparisons of full configured conditions.

**Parser and contract diagnostics.** Llama direct and Llama CoT have 121 and 122 first-stage parser rejections across all families, caused by completed outputs that do not satisfy the frozen final-JSON extraction rule. In the direct condition, 120 of these contain a fenced JSON object surrounded by extra prose. When configurations commit to numerical predictions, most values fall outside tolerance: 35 of 40 for Qwen3-8B direct, 28 of 31 for AWQ-off, and 16 of 17 and 19 of 22 for Gemma direct and CoT. These outputs are not repaired for the primary score. Qwen3-8B thinking and AWQ-on abstain on all predictive items. Thus zero predictive success for those conditions is not the same phenomenon as 40 wrong numerical predictions. Table 6 decomposes predictive and counterfactual outcomes for all ENRICHED conditions.

For the Gemma and DeepSeek conditions, generation completion and parseability are reported separately in Table 21. Gemma's length-limited outputs fall mainly on explanation items (11 per configuration) and on seven Gemma-direct predic-

| Condition | Backend | Precision | Temp. | Top-$p$ | Top-$k$ | Max out | Timeout | Thinking |
|---|---|---|---|---|---|---|---|---|
| Qwen3-8B direct | Transformers | bf16 / none | 0.7 | 0.80 | 20 | 1,024 | 120 s | off |
| Qwen3-8B thinking | Transformers | bf16 / none | 0.6 | 0.95 | 20 | 8,192 | 360 s | on |
| Llama-3.1-8B direct | Transformers | bf16 / none | 0.6 | 0.90 | 50 | 1,024 | 120 s | – |
| Llama-3.1-8B CoT | Transformers | bf16 / none | 0.6 | 0.90 | 50 | 4,096 | 240 s | – |
| Qwen3-32B-AWQ off | vLLM | fp16 / AWQ | 0.6 | 0.90 | 20 | 1,024 | 120 s | off |
| Qwen3-32B-AWQ on | vLLM | fp16 / AWQ | 0.6 | 0.95 | 20 | 8,192 | 360 s | on |
| Gemma-3-27B QAT direct | llama.cpp | Q4_0 / f16 KV | 0.6 | 0.90 | 50 | 1,024 | 120 s | – |
| Gemma-3-27B QAT CoT | llama.cpp | Q4_0 / f16 KV | 0.6 | 0.90 | 50 | 4,096 | 240 s | – |
| DeepSeek-R1-Distill-Qwen-32B | vLLM | bf16 / NF4 | 0.6 | 0.95 | 20 | 8,192 | 360 s | – |

Table 18: Frozen answering settings. Precision gives the compute precision and the quantization or KV-cache format; Thinking gives the native-thinking switch where one exists. Chat templates are rendered with `add_generation_prompt=True`; Qwen additionally receives the frozen `enable_thinking` flag. No custom stop strings are supplied.

| Condition | Input | Direct | Predictive | Counterfact. | Expl. facts | Resolved clar. |
|---|---|---|---|---|---|---|
| Qwen3-8B direct | Enriched | 20.0 [10.0,30.0] | 12.5 [5.0,22.5] | 35.0 [27.5,45.0] | 62.5 [52.5,70.0] | 10.0 [2.5,20.0] |
| Qwen3-8B thinking | Enriched | 52.5 [40.0,65.0] | 0.0 [0.0,0.0] | 12.5 [5.0,22.5] | 80.0 [70.0,90.0] | 25.0 [15.0,35.0] |
| Llama-3.1-8B direct | Enriched | 0.0 [0.0,0.0] | 0.0 [0.0,0.0] | 17.5 [10.0,25.0] | 15.0 [5.0,25.0] | 0.0 [0.0,0.0] |
| Llama-3.1-8B CoT | Enriched | 0.0 [0.0,0.0] | 0.0 [0.0,0.0] | 0.0 [0.0,0.0] | 30.0 [17.5,42.5] | 0.0 [0.0,0.0] |
| Qwen3-32B-AWQ off | Enriched | 72.5 [67.5,75.0] | 7.5 [0.0,15.0] | 7.5 [0.0,15.0] | 65.0 [57.5,72.5] | 32.5 [25.0,40.0] |
| Qwen3-32B-AWQ on | Enriched | 92.5 [85.0,100.0] | 0.0 [0.0,0.0] | 17.5 [9.9,27.5] | 52.5 [37.5,65.0] | 30.0 [22.5,40.0] |
| Gemma-3-27B direct | Enriched | 0.0 [0.0,0.0] | 2.5 [0.0,7.5] | 0.0 [0.0,0.0] | 2.5 [0.0,7.5] | 0.0 [0.0,0.0] |
| Gemma-3-27B CoT | Enriched | 0.0 [0.0,0.0] | 7.5 [0.0,15.0] | 10.0 [2.5,17.5] | 2.5 [0.0,7.5] | 0.0 [0.0,0.0] |
| DeepSeek-R1-32B | Enriched | 60.0 [45.0,75.0] | 0.0 [0.0,0.0] | 27.5 [17.5,37.5] | 27.5 [15.0,40.0] | 0.0 [0.0,0.0] |
| Qwen3-8B direct | Struct.-only, recovered | 0.0 [0.0,0.0] | 12.5 [5.0,22.5] | 35.0 [27.5,45.0] | 65.0 [57.5,72.5] | 0.0 [0.0,0.0] |
| Qwen3-32B-AWQ on | Struct.-only | 97.5 [92.5,100.0] | 2.5 [0.0,7.5] | 15.0 [7.5,22.5] | 62.5 [50.0,72.5] | 32.5 [22.5,42.5] |

Table 19: Fixed-denominator endpoints for all conditions on the common 200-item core, with nominal 95% paired whole-trajectory bootstrap intervals (10,000 replicates, stratified by domain and family). Values are percentages over 40 items per family. The Gemma and DeepSeek conditions have no structured-only ablations, so Tables 7 and 20 remain limited to the two ablated conditions.

tive items. On direct readout, all 40 answers of each Gemma configuration are valid JSON but violate the typed answer contract. DeepSeek completes every initial response, and 29 of its explanation responses are rejected by the frozen final-JSON parser.

**Clarification diagnostics.** Structured-only Qwen3-8B produces schema-valid first turns on all 40 items: 35 route correctly, and 5 request a wrong supported slot. None of its 35 continuations is correct; 11 violate the resolved schema, and 24 are valid but wrong. Every Qwen and Llama ENRICHED configuration declares a clarification on all 40 items, whereas Gemma direct, Gemma CoT, and DeepSeek do so on 1, 2, and 3 items (2.5–7.5%). Gemma direct routes no item. Gemma CoT routes one: its structured selector correctly requests the execution condition while its readable question asks for a unit, and automatic routing follows the structured selector. Its continuation answers a 1-s query with the observed 0.2-s angle, which is incorrect. DeepSeek routes one item, but its attempted continuation saved no answer.

**Restart recovery.** The Qwen3-8B structured-only run originally lost 35 initial outputs to an external restart. After an authorized recovery with identical requests and settings, its apparent explanation advantage for enriched input disappeared. All reported values for this condition are the recovered values.

## K Resource-evaluation evidence and human review

The resource evaluation asks whether SimPhysQA provides the components needed to build and evaluate AI systems over the five families.

**Prepared cohorts.** The original admitted predictive source contains 11,105 QA records across 6,340 trajectory groups. The prepared enriched union is bounded: 56 development items, 140 annotation-audit items, and a 400-item held-out pool that contains the nested 200-item core. The 200-item core is the evaluated subset for every

| Comparison | Family | Difference and interval | Interpretation |
|---|---|---|---|
| Qwen3-8B structured-only minus enriched | Direct | -20.0 [-30.0, -10.0] | Enriched input helps direct readout for this condition. |
| Qwen3-8B structured-only minus enriched | Predictive | 0.0 [0.0, 0.0] | No observed difference. |
| Qwen3-8B structured-only minus enriched | Counterfactual | 0.0 [0.0, 0.0] | Recovered data eliminate the original apparent enriched advantage. |
| Qwen3-8B structured-only minus enriched | Explanation facts | +2.5 [0.0, +7.5] | Small structured-only point advantage after recovery. |
| Qwen3-8B structured-only minus enriched | Clarification | -10.0 [-20.0, -2.5] | Enriched input helps resolved clarification for this condition. |
| AWQ-on structured-only minus enriched | Direct | +5.0 [-2.5, +12.5] | No clear enriched advantage. |
| AWQ-on structured-only minus enriched | Predictive | +2.5 [0.0, +7.5] | Small point difference, interval touches zero. |
| AWQ-on structured-only minus enriched | Counterfactual | -2.5 [-15.0, +10.0] | No clear difference. |
| AWQ-on structured-only minus enriched | Explanation facts | +10.0 [-7.5, +27.5] | No clear difference. |
| AWQ-on structured-only minus enriched | Clarification | +2.5 [-5.0, +12.5] | No clear difference. |

Table 20: Matched representation ablations. Negative values favor enriched input. The ablations are available only for Qwen3-8B direct and Qwen3-32B-AWQ-on.

| Condition | Completed | Length-limited | Completed & parseable |
|---|---|---|---|
| Gemma-3-27B direct | 181/200 | 19/200 | 139/200 |
| Gemma-3-27B CoT | 189/200 | 11/200 | 151/200 |
| DeepSeek-R1-32B | 200/200 | 0/200 | 134/200 |

Table 21: Initial-response execution status for the Gemma and DeepSeek conditions. All selected items stay in the primary denominators.

reported model result; the remaining pool items are part of the dataset inventory. The prepared union contains 596 QA items across 556 trajectory groups. These counts are included for provenance but should not displace the main contribution narrative.

**Family coverage.** The prepared union contains 124 direct, 124 predictive, and 116 each of counterfactual, explanation, and clarification records. The nested main core contains 40 items per family, 50 per domain, and 10 items per domain-family cell.

**Why this is a resource rather than a task collection.** The contribution is not only a set of question strings. Each item packages a public task view, a private verification view, an answer contract, evidence spans, and family-specific scoring material. This design makes the same trajectory reusable across different scientific questions without leaking private outcomes into the model input. It also makes negative results interpretable: an incorrect prediction, an abstention, a unit error, and a failed clarification route are different observable failures rather than one undifferentiated wrong answer. In this sense, SIMPHYSQA is closer to an evaluation substrate than to a static QA list. The current release is intentionally bounded, but it establishes the schema, evidence discipline, validation checks, audit protocol, and execution archive needed for extension.

**Annotation audit: design.** Three reviewers independently assessed the same 140 annotation packets: four domains $\times$ five families $\times$ seven packets, for 420 individual ratings. There was no separate calibration set, and original ratings were submitted before any discussion. Each packet receives one primary judgment, *acceptable* or *material_error*, within a focused scope of five criteria: physical/global context, trajectory language, question wording and type, timestamp language, and

state–event descriptions. A material error is one that could change the physical interpretation, the target entity, time, unit, or frame, or the intended question, or one that asserts an unsupported fact or effect. Minor awkward wording is recorded as a comment and does not make a packet unacceptable. For each material error, reviewers record the criterion, an error category, a source location, and a comment. The complete frozen reviewer guide is reproduced below.

**Annotation audit: results.** Table 22 reports the share of the 140 packets that each reviewer judged acceptable. The mean of the three individual rates is approximately 90%. These are rates of individual ratings. They are neither the share of packets accepted unanimously nor a measure of inter-rater agreement, which we do not report here.

| Reviewer | Acceptable | Material error |
|---|---|---|
| R1 | 100% | 0% |
| R2 | 83% | 17% |
| R3 | 87% | 13% |
| Mean of individual rates | ≈90% | ≈10% |

Table 22: Focused annotation audit: per-reviewer share of the same 140 packets judged acceptable. These are rates of individual ratings, not unanimous packet acceptance or inter-rater agreement.

**Annotation audit: scope.** The audit assesses the model-facing annotation layer. Gold answers, explanations, and private traces may be consulted to corroborate this scope, but the audit does not establish the correctness of every reference answer, verification trace, or model explanation, and it does not rate answering-model performance. With seven packets per domain–family cell, it does not support per-cell estimates. Discussion, adjudication, and any proposed corrections are kept separate from the frozen packets and original ratings.

```
# Focused reviewer guide /1.0.0

Review independently using your assigned R1, R2 or
      R3 slot. All three reviewers
inspect the same 140 items; slots do not specify
     personal identities. No separate
calibration set. Copy your frozen CSV form before
     editing. Submit original ratings
before discussion. Blank judgments are uncollected,
      never acceptable ratings.

Primary judgment: acceptable or material_error **
     within the focused scope**:
physical/global context, trajectory language,
     question wording/type, generated
timestamp language and state-event descriptions. A
      material error could change
the physical interpretation, target/entity/time/
     unit/frame or intended question,
or asserts an unsupported fact/effect. Optional
     gold answers, explanations, traces
and extra private verification corroborate this
     scope; they are not automatically
part of a comprehensive answer-quality audit. Do
     not rate answering-model performance.

Supply each criterion as acceptable/material_error
     /not_applicable:
global_context; trajectory_language;
     question_type_wording; timestamp_language;
state_event_descriptions. Use not_applicable only
     for a criterion that has no
applicable content (e.g. no declared events);
     missing required content is an error.
Global context describes physical setup;
     trajectory text describes the observed
window. They need not be duplicate paraphrases.
     Timestamp text is generated
language joined by state_id, distinct from the
     generic original state.context.

Error categories: factual_contradiction,
     unsupported_claim, wrong_entity,
wrong_time, wrong_unit, wrong_frame,
     misleading_event_semantics,
question_type_mismatch, readability,
     missing_required_field, other. Minor awkward
wording may be recorded in comments while the item
      remains acceptable. Use the
readability error category for material ambiguity/
     illegibility. For material_error
record at least one material criterion, category,
     source location (JSON pointer,
state/event/link ID or question field) and an
     explanatory comment. Separate
multiple categories/locations with semicolons.
     Criterion CSV syntax:
`global_context=acceptable;trajectory_language=
     acceptable;question_type_wording=acceptable;
     timestamp_language=acceptable;
     state_event_descriptions=not_applicable`.

Assess statements against original measurements/
     relations, entities and declarations,
with their units/frames and exact temporal
     relationships. Scheduled future events
are declarations, not evidence they were applied
     or caused an effect. Ended timing
does not mean effects ended; torque/force limits
     do not establish realized values.
Shared event text repeats per state with that
     state's actual relationship.
Explicitly labeled unresolved semantics or
     limitations are acceptable if the prose
faithfully conveys what is established and
     uncertain. Asserted resolution without
evidence, concealed uncertainty that changes
     interpretation, or contradicted source
is material_error. Never convert a declared
     limitation into an invented fact.

Direct asks an observed readout. Predictive asks
     future outcomes. Counterfactual
asks the verified omission world; a Boolean can
     remain unchanged and a numerical
contrast can be within tolerance. Explanation
     intentionally supplies its selected
comparison outcomes in a separate task field, not
     in observed states; assess
question/family framing and public descriptions,
     not automatically every private
gold claim. Clarification is an explicit absent
     query selector with two compatible
incompatible-answer completions. Inspect labeled
     initial question/view, authored
clarification, recorded author reply and resolved
     question/view. They are reference
turns, not an executed model conversation. The
     reply supplies information, not gold.

Freeze original submissions immutably. Discussion,
      adjudication, proposed annotation
corrections and any later rubric revision live
     separately and do not change the
packets or original independent ratings. Do not
     begin formal review from the
development extraction-check index or header-only
     preparation templates.
```

**Model-answer prose study.** The prospective model-answer prose study freezes eight explanation tasks, two per domain, for all nine enriched conditions, giving 72 planned responses. All 72 now exist: 48 from the six Qwen and Llama conditions and 24 from the Gemma and DeepSeek conditions. Six of the latter are length-limited (two from Gemma direct, four from Gemma CoT) and remain in the subset unchanged. Human assessment of model explanation prose is a separate study whose two independent blinded assessor forms remain uncollected, so the paper reports no human prose-adequacy result.

## L Responsible NLP checklist mapping

This section maps the paper to common Responsible NLP review concerns.

**Data and documentation.** The manuscript identifies the physical domains, QA families, public/private split, validation checks, release scope, and missing evidence boundaries. It distinguishes the original predictive inventory from the prepared five-family enriched union.

**Annotation.** The paper describes automatic enrichment, reproduces every executed generation, repair, and evaluation prompt, and describes the focused annotation audit protocol. It reports per-reviewer acceptance rates from the focused annotation audit without presenting them as inter-rater agreement, and it does not claim human prose-quality judgments.

**Models.** The paper lists all nine evaluated conditions (five models), runtime classes, reasoning controls, generation settings, execution status, and model artifacts.

**Evaluation.** The paper reports fixed denominators, missingness handling, parsing diagnostics, typed scoring, clarification routing, explanation factual scoring, and bootstrap intervals. It explicitly avoids pooling incompatible endpoints into one headline score.

**Compute.** The paper reports serial local execution on two workstations, their verified runtimes, and frozen generation settings. It does not claim measured GPU utilization.

**Risks and intended use.** The resource is intended for research on grounded scientific QA and evaluation methodology. It should not be used as evidence that a model is safe for real-world robotic, vehicle, or physical control deployment.

**Open limitations.** The main open evidence gaps are independent model-answer prose review, exact per-reviewer counts and agreement statistics for the annotation audit, full-release enrichment, and broader physical-domain coverage.

## M Readable examples for all five families

This section gives schematic examples of how the same kind of trajectory evidence becomes different QA tasks. The examples are deliberately written in ordinary language; exact item text in the dataset is preserved in the frozen request files.

### M.1 Direct readout example

**Public evidence.** The public record shows a tethered-load trajectory up to time $t = 0.4$ s. One public state contains the load swing angle with entity `load`, quantity `swing_angle`, unit radians, frame `world`, and the exact timestamp.

**Question.** What is the load swing angle at $t = 0.4$ s?

**Expected behavior.** The model should locate the public state, return the target ID and a numeric value with the declared unit and frame, and not invent a different time or object.

**Scoring.** The answer is correct only if the JSON target is valid, the unit/frame match, and the scalar value is within the target tolerance.

### M.2 Predictive example

**Public evidence.** The model sees vehicle states up to the observation boundary and the declared braking controller. The future terminal state is not public.

**Question.** What will the car's forward velocity be at $t = 2.0$ s?

**Expected behavior.** The model must answer from the visible prefix and declared controls. It cannot read the future value from the prompt because that value is private.

**Scoring.** The prediction is compared against the private admitted reference outcome. Abstention is counted as unsuccessful in the primary endpoint.

### M.3 Counterfactual example

**Public evidence.** The model sees the factual scenario and a task statement that disables a scheduled intervention, such as omitting a push or braking command. It does not see the alternative rollout.

**Question.** If the scheduled push were omitted, would block 2 remain supported at the target time?

**Expected behavior.** The model answers for the specified alternative condition, not the original factual trajectory. The answer may be unchanged relative to the factual world; unchanged outcomes remain legitimate.

**Scoring.** The answer is compared against the verified matched alternative. The factual replay and alternative admission evidence remain private.

### M.4 Explanation example

**Public evidence.** The model is intentionally shown two selected readouts: the factual scheduled outcome and the matched omission outcome. The prompt states which intervention changes and which conditions remain fixed.

**Question.** Explain the comparison between the factual and omission outcomes.

**Expected behavior.** The model should identify the intervention, fixed conditions, two readouts, whether they are exactly equal, whether the existing scorer distinguishes them, and a careful interpretation with uncertainty.

**Scoring.** The automatic endpoint checks structured facts only. A separate human study is required for prose adequacy.

### M.5 Clarification example

**Public evidence.** The initial public view leaves a selector unresolved. For example, a question may not specify which object, time, or execution condition is intended.

**Question.** Did it succeed?

**Expected behavior.** The model should not guess. It should ask for the missing selector. After receiving the recorded reply, it should answer the resolved task.

**Scoring.** Stage 1 is judged by structured gap recognition and slot match. Stage 2 is judged by typed answer correctness. The stages are reported separately.

## N Prompt and serialization discipline

**Why exact serialization matters.** The same task can become easier, harder, or ambiguous if fields are reordered, omitted, paraphrased, or accidentally merged. SimPhysQA therefore preserves exact rendered requests, model chat templates, tokenization checks, raw responses, and parser/scorer versions. The paper's tables summarize this evidence; the execution package stores the exact request bytes.

**Public request assembly.** Each public request has a condition identifier, a sample identifier, no tools, and a message containing instructions plus the canonical public task view. Direct/predictive/counterfactual requests append the typed-target response contract. Explanation requests use the explanation-specific response contract. Clarification requests use the initial ambiguous public view and a schema for requesting missing information.

**Structured-only projection.** The structured-only ablation removes derived trajectory, timestamp, and event descriptions. It preserves the underlying task facts, original context, controls, question, answer specification, supplied explanation comparisons, and clarification ambiguity. It is therefore an input-representation ablation, not a removal of answerability.

**No answer repair.** The primary evaluation does not repair outputs, substitute gold turns, reopen failed answers, or score alternate parser interpretations. Diagnostic reports may classify rejected output surfaces, but those classifications do not change the primary score.

## O Additional interpretation notes for readers

**A high direct score is not physical understanding.** Direct QA often tests whether the model can locate visible evidence and return it in the required contract. This is useful, but it does not imply future prediction or counterfactual reasoning.

**A high explanation-fact score is not prose adequacy.** The automatic explanation endpoint is intentionally narrow. It checks whether the model reports the supplied comparison facts faithfully. It does not judge whether the model gives a convincing, causally careful, or pedagogically useful explanation.

| Artifact | Preserved information | Reason |
|---|---|---|
| Prepared request | Exact public prompt, condition ID, input representation, response contract | Ensures the benchmark can be rerun and audited. |
| Token/render check | Model-specific template and token counts | Detects context overflow or serialization differences. |
| Raw output | Complete returned text and termination evidence | Prevents silent cleanup or selective parsing. |
| Parsed answer | Final JSON object or parser failure | Separates surface format from scientific correctness. |
| Scoring row | Typed validity, target correctness, errors, status | Makes denominator and failure policy inspectable. |
| Timing/memory receipt | Loading, generation, process lifecycle, memory samples where available | Documents execution feasibility without overclaiming GPU utilization. |

Table 23: Execution artifacts that support reproducibility and error analysis.

**Clarification has two failures.** A model can fail by answering too early, asking for the wrong slot, or failing to answer after receiving the correct selector. SimPhysQA reports these pathways separately because they correspond to different assistant behaviors.

**Low predictive success is not a general inability.** Predictive and counterfactual scores measure end-to-end success under the supplied public evidence and a tool-free interface. Abstentions, output-contract failures, and incorrect values are reported separately (Table 6). Because public-input sufficiency has not been independently established and no computational tools are available, low scores do not separate reasoning difficulty from information limits or the absence of numerical integration.

**Ablation scope is limited.** Only two structured-only ablations are completed. This is enough to test whether language enrichment helps those conditions, but not enough to claim a universal input-representation effect. The paper therefore reports selective evidence rather than a broad enrichment-benefit claim.

**Mixed results are expected for a useful resource.** A resource that exposes different capabilities should not necessarily produce a clean monotonic leaderboard. The value of SimPhysQA is that it can show when reading, predicting, explaining, and clarifying diverge.

## P What the release supports and does not support

**Supported.** The current evidence supports the claim that SimPhysQA provides a bounded, five-family, simulator-grounded QA resource with enriched trajectory records, public/private separation, family-specific evidence contracts, and local-model results for nine enriched conditions of five models and two structured-only ablations on a common 200-item core.

**Not supported yet.** The current evidence does not support claiming full enrichment of every original predictive record, human explanation-quality conclusions, semantic entailment of every generated sentence, sufficiency of the public inputs to determine every future target, real-world physical reliability, broad out-of-distribution dialogue competence, or a universal language-enrichment benefit.

**Why this honesty strengthens the paper.** A resource paper is judged not only by positive results but by whether its evidence boundaries are clear enough for other researchers to trust and reuse. The paper therefore treats unavailable or pending evidence as part of the scientific record rather than smoothing it away.